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GazeLLM: a plug-and-play zero-shot LLM reasoning framework for boosting gaze target detection

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Abstract

In the gaze behavior understanding task, existing vision-based models demonstrate inherent limitations in high-dimensional semantic understanding, while vision-language models (VLMs) encounter challenges in precise object localization. To address this issue, we propose GazeLLM, the first zero-shot large language model (LLM) boosted framework for gaze target reasoning. Our key innovations include three aspects. First, we have structured object extraction. Using off-the-shelf detectors (e.g., MM-GroundingDINO and Depth Anything V2), we convert images into 3D object representations, including head and gaze direction, object categories, and metric depth. Second, we implemented an autonomous chain-of-thought (CoT) reasoning system. We designed self-generated CoT prompts to guide pretrained LLMs, such as ChatGPT o3-mini-high, to predict gaze targets via spatial-semantic analysis. Third, we proposed a plug-and-play module. We employed a novel cross-modal fusion mechanism that combines the LLM's probability dictionaries with vision-based gaze heatmaps via Gaussian-weighted multi-hot mapping. Extensive experiments show that GazeLLM significantly improves state-of-the-art models, increasing their performance from 17% to 34% on challenging cases, such as long-range targets or rare categories, without the need for retraining. It also extends seamlessly to multi-person social gaze tasks (e.g., a 42% LAEO AP gain on the AVA-LAEO benchmark). Our framework demonstrates superior generalizability and interpretability compared to VLMs, validating the efficacy of LLMs in understanding gaze behavior by mining semantic cues.

Keywords: Gaze target detection, Social gaze prediction, Large language model (LLM), Prompt engineering

1 Introduction

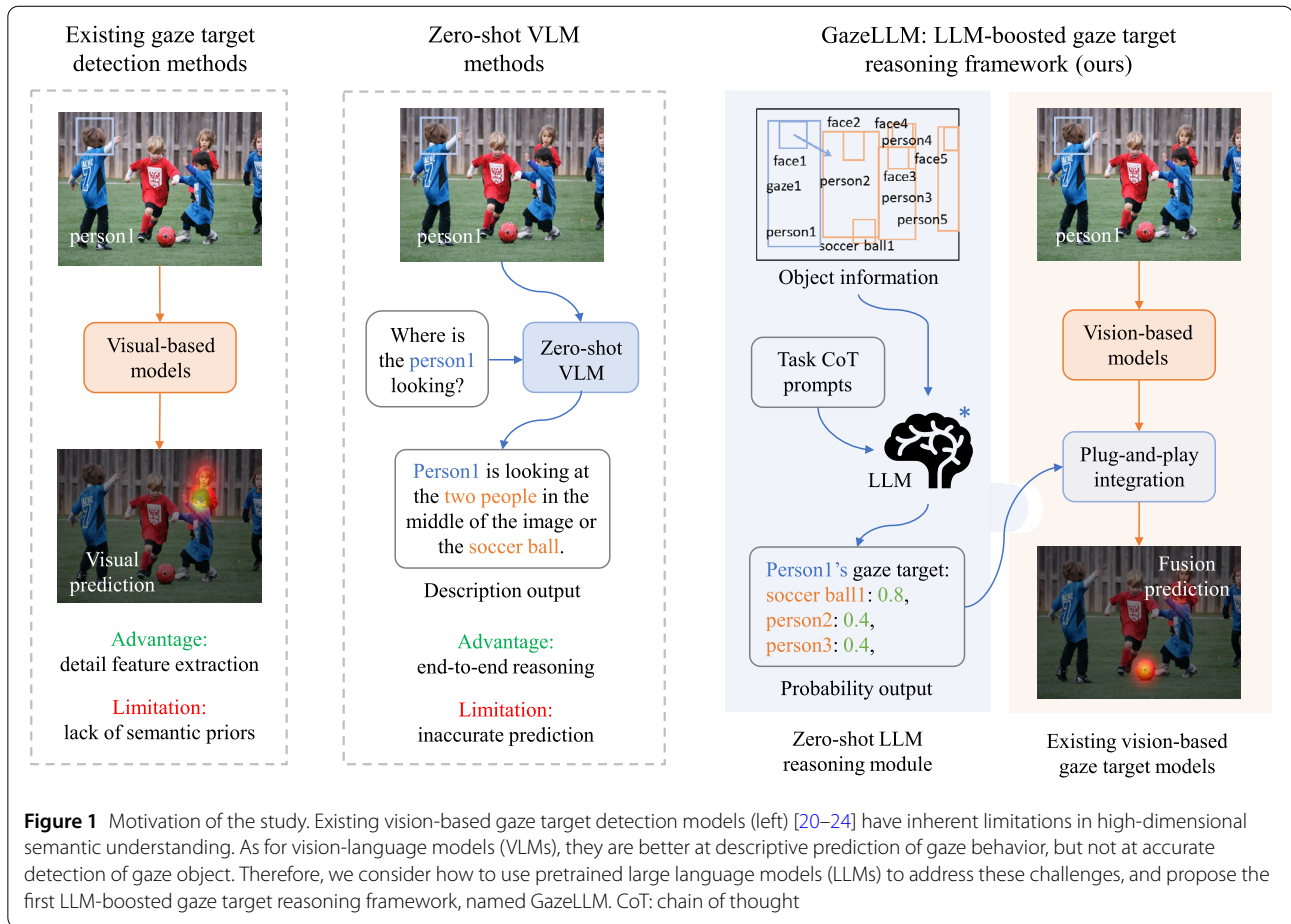
Gaze behavior understanding is an important field in computer vision, as it is highly valuable for analyzing human visual attention [1–4]. Initially, the community proposed the gaze estimation task, which predicts the three-dimensional (3D) gaze direction from a cropped image of a human head [5–9]. Subsequently, the scope of this task was expanded to encompass gaze target detection, a process that localizes the specific point or object being viewed in an unconstrained image [10–13]. The latter aspect is the focus of this study. Besides, gaze target detection has been

expanded to encompass multi-person social gaze prediction, including shared attention detection (SA) [14], which predicts the shared gaze target of a group of people, and look-at-each-other detection (LAEO) [15], which distinguishes whether two people are looking at each other.

Recent advances in deep learning have substantially improved gaze target detection accuracy through sophisticated architectures such as multi-scale attention mechanisms and 3D scene understanding [16–19]. However, existing vision-based gaze target detection models [20–24] have inherent limitations in high-dimensional semantic understanding. As shown in Fig. 1 (left), they focus on low-level visual pattern matching between head orientation and potential targets, overlooking the high-level semantic relationships that govern human attention allocation.

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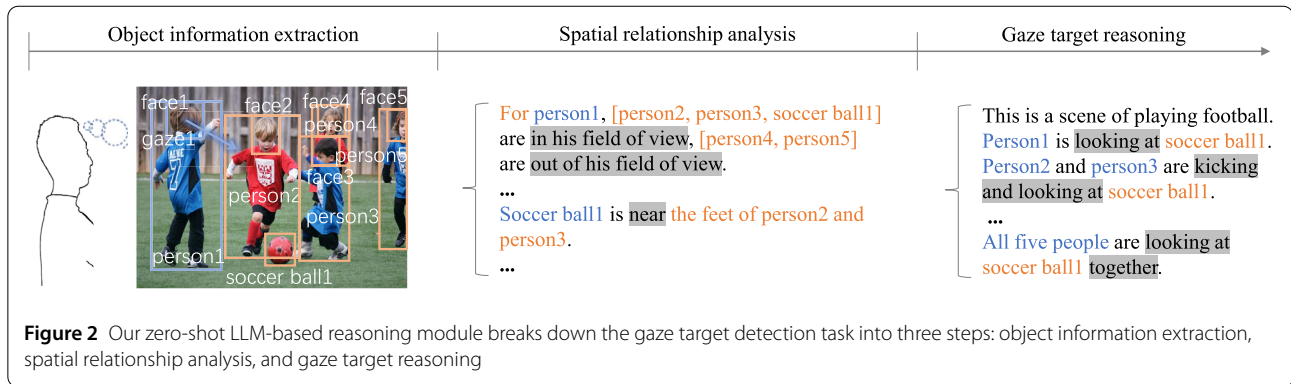
tion. For instance, a tennis player's gaze naturally follows the ball trajectory due to task demands, yet current models fail to leverage such semantic prior.

We consider how to alleviate the inherent limitations of existing vision-based models in high-dimensional semantic understanding in gaze target detection. As shown in Fig. 1 (middle), one approach to addressing this challenge is to introduce advanced vision-language models (VLMs) [25–34]. However, extensive experiments demonstrate that existing pretrained VLMs are better at descriptive prediction of gaze behavior, but not at accurate detection of gaze object (see Sect. 4.5). In addition, although fine-tuning VLMs on this task can effectively improve their performance, it will obviously bring huge training costs.

Therefore, we consider how to use pretrained large language models (LLMs) [35–42] to address these challenges. Through large-scale natural language pretraining, LLMs contain rich prior knowledge and powerful reasoning ability. However, applying text-only LLMs to gaze target detection still requires addressing the issue of their inability to read image inputs. Recently, multiple works [43–46] choose to directly input key text information ex-

tracted from images, rather than the images themselves, into text-only LLMs for reasoning complex visual tasks. This approach allows for leveraging the powerful chain-of-thought (CoT) reasoning ability of pretrained LLMs, while eliminating the interference of task-irrelevant visual information present in images. The key to these approaches is determining what kind of text information to extract as input for text-only LLMs, and designing appropriate CoT reasoning prompts for specific visual tasks.

As shown in Fig. 1 (right), we design a zero-shot LLM-based reasoning module for gaze target detection. Our module breaks down this task into three steps: object information extraction, spatial relationship analysis, and gaze target reasoning (Fig. 2). First, we convert the original image into structured 3D object representations, including human head location and gaze direction, object category and location, as well as metric depth information. Then, we implement appropriate CoT prompting strategies to guide the pretrained LLM to predict possible gaze targets step by step from the spatial relationship between humans and objects. Next, we design a simple cross-modal integration mechanism. This mechanism allows our zero-shot LLM reasoning module to seamlessly integrate with



existing vision-based gaze target models in a plug-and-play manner [47], effectively compensating for their limitations in high-dimensional semantic understanding. Overall, the above steps constitute our complete LLM-boosted gaze target reasoning framework, named GazeLLM.

With the help of our reasoning framework, existing vision-based gaze target models generally achieve a performance improvement of 17% to 34% on challenging cases, such as long-range and rare-category gaze targets, without the need for retraining (see Sect. 4.1). By simply adjusting the task prompts, GazeLLM can be easily extended to downstream multi-person social gaze tasks and help existing vision-based models achieve state-of-the-art (SOTA) performance (see Sect. 4.2). Despite the absence of image inputs, our LLM-based zero-shot reasoning module still outperforms advanced VLMs in gaze target detection, validating both the efficacy and rationality (see Sect. 4.5).

Our main contributions are as follows:

- 1) We propose the first LLM-boosted gaze target reasoning framework GazeLLM. In GazeLLM, we design a zero-shot LLM reasoning module that can accurately predict possible gaze targets from the extracted textual object information.
- 2) Our zero-shot LLM reasoning module can be seamlessly integrated with existing vision-based gaze target models in a plug-and-play manner, effectively compensating for their limitations in high-dimensional semantic understanding.
- 3) Our GazeLLM framework can be easily extended to downstream multi-person social gaze tasks, exhibiting greater flexibility and generalization compared to traditional vision models.

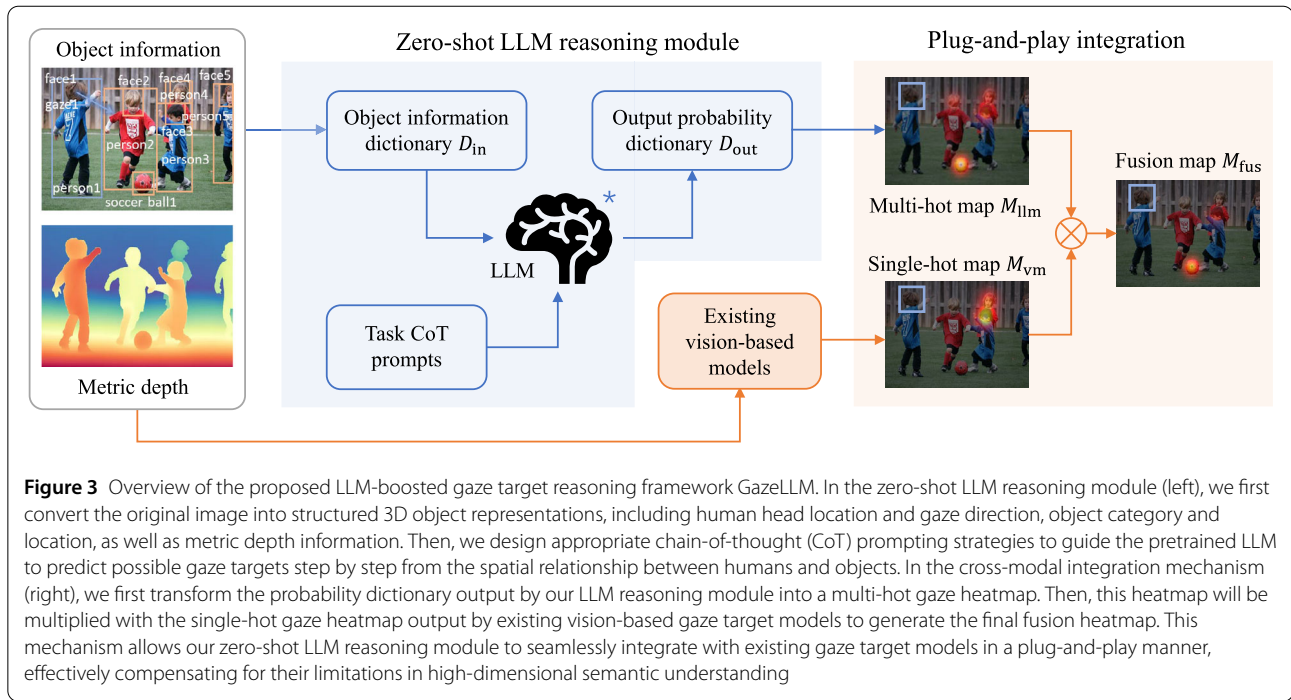
2 Related work

Recasens et al. [10] pioneered the gaze target detection task and introduced the GazeFollow dataset, comprising a substantial collection of images annotated with head positions and corresponding gaze targets. Lian et al. [11] augmented viewpoint supervision using a multi-scale field of view attention mechanism. Chong et al. [13] extended the

task by introducing a video dataset VideoAttn and addressing out-of-frame scenarios. Fang et al. [16] integrated monocular depth as a geometric prior. Bao et al. [17] leveraged 3D geometric constraints to refine accuracy, and Yang et al. [48] incorporated human-object interaction cues for enhanced context. Gupta et al. [23] jointly modeled gaze targets and social gaze patterns. Yang and Lu [49] introduced an additional local view that is more focused on the gaze target itself. Tafasca et al. [22] unified gaze-target localization and semantic labeling. Recently, Song et al. [50] streamlined pipelines by leveraging foundational vision encoders (ViTs). Tu et al. [51] redefined gaze analysis by proposing human-gaze-target detection, simultaneously localizing all $\langle \text{human head}, \text{gaze target} \rangle$ pairs in an image through an end-to-end DETR-based paradigm. Tonini et al. [52] improved head-target associations but retained dependence on preliminary object detection, which limits their model's generalization. Tu et al. [53] unified gaze location and gaze object detection within an enhanced transformer architecture.

As for multi-person social gaze prediction, Fan et al. [14] proposed the shared attention detection task, which predicts the shared gaze target of a group of people. Marin-Jimenez et al. [15] proposed the look-at-each-other detection task, which distinguishes whether two people are looking at each other. Besides, Tafasca et al. [20] proposed the look-at-head detection task. Recently, Gupta et al. [23] proposed a unified framework MTGS, which can simultaneously perform gaze target detection and social gaze prediction.

Large language models (LLMs) [35–41, 54–57] transform natural language processing by demonstrating powerful abilities in language understanding and generation. GPT-3 [56] introduced a large-scale auto-regressive model that excels at few-shot learning across various tasks. GPT-4 [35] extended these capabilities with a larger model architecture and improved handling of complex reasoning tasks. Subsequent advancements introduced efficiency-focused paradigms: DeepSeek-R1 [41] pioneered hybrid sparse-dense attention for real-time inference, while



ChatGPT-o3-mini [42] achieved mobile deployment via extreme quantization without obvious performance degradation.

While conventional LLMs are designed for text-only processing, recent studies explored their adaptation to visual tasks through innovative prompting strategies [58–60]. Multiple approaches converted visual inputs into textual descriptions using external vision models. CoOp [61] extended prompt engineering by learning task-specific prompts for visual tasks, improving performance on unseen categories. Img2LLM [43] proposed a plug-and-play module that provides LLM prompts to enable LLMs to perform zero-shot VQA tasks without end-to-end training. Visual ChatGPT [44] used object detectors to enable LLMs to reason about visual content through language proxies. GPT-DRIVER [45] represented the planner inputs and outputs as language tokens, and leveraged the LLM to generate driving trajectories through a language description of coordinate positions. Symbol-LLM [46] extracted the visual content of the images and checked as symbols and used LLMs to reason out activity semantics based on rules via fuzzy logic calculation.

Inspired by these works, we attempt to use text-only LLMs to mine semantic cues for gaze target detection and multi-person social gaze prediction without any additional training or fine-tuning.

3 GazeLLM framework

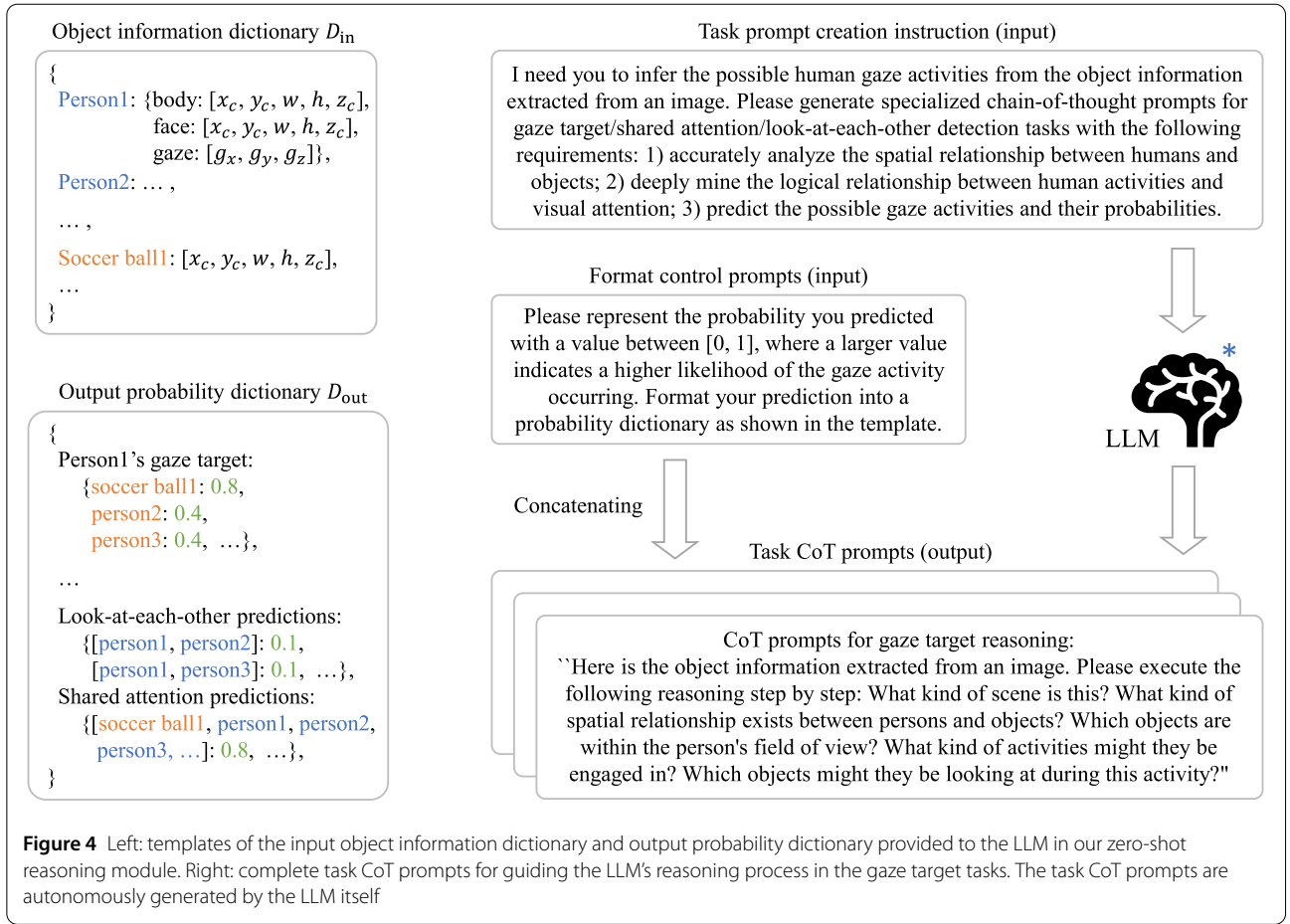
Figure 3 is an overall presentation of the proposed LLM-boosted gaze target reasoning framework GazeLLM. Sec-

tion 3.1 introduces our zero-shot LLM reasoning module that can accurately predict possible gaze targets from the extracted textual object information. Section 3.2 introduces our cross-modal integration mechanism that can seamlessly integrate our LLM reasoning module with existing vision-based gaze target models in a plug-and-play manner.

3.1 Zero-shot LLM reasoning module

Object information extraction. The first step of our method is to consider what object information needs to be extracted and how to extract it from the original image. The complete human gaze behavior includes the human head location and gaze direction, as well as the gaze target location and category. Furthermore, since the real world is 3D, human gaze behaviors also include the 3D spatial relationship between a person and the target which the person is looking at. Therefore, for an input image, we use the advanced detector MM-GroundingDINO [62] to detect all LVIS-category objects [63] and use GazeOnce [64] to estimate all people’s 3D gaze direction. Besides, we also use Depth Anything V2 [65] to estimate the metric depth information. Please note that, the gaze estimation module and depth estimation module are also commonly used in existing vision-based gaze target methods [12, 13, 16–21, 23]. That is to say, the object information extraction process in our module will not impose excessive computational burden on existing methods.

Object information formatting. To facilitate the LLM’s understanding of extracted object information and en-



able batch processing, we format these object information into a dictionary. Specifically, we define the detected human instances as $\{o_1, o_2, \dots, o_M\}$ and the detected object instances as $\{o_{M+1}, o_{M+2}, \dots, o_N\}$. In this way, all human and object instances are represented as $\{o_i \mid i = 1, 2, \dots, N\}$. Then, as shown in Fig. 4 (left), we create a formatted dictionary D_{in} , where each key-value pair stores the 3D location of a human/object instance. Among them, the value corresponding to each human is a dictionary that includes their body location, head location, and 3D gaze direction $[g_x, g_y, g_z]$. All the above 3D location information is composed of the 2D bounding box coordinates $[x_c, y_c, w, h]$ and the metric depth coordinates of the center point z_c . For the batch of input images $I_k, k = 1, 2, \dots, K$, we obtain a batch of corresponding object information dictionaries $D_{in}^k, k = 1, 2, \dots, K$, which will serve as the text input for LLMs. In addition, to facilitate the LLM's understanding of extracted object information, we also provide it with an object dictionary template.

Task CoT prompts. With the extracted object information, we need to provide the LLM with task prompts to guide its reasoning process in gaze target detection. Numerous studies [66–70] prove that guiding LLMs step by

step through chain-of-thought (CoT) prompts for logically complex tasks often results in higher accuracy than letting LLMs output the final result directly. Therefore, it is necessary to design appropriate CoT prompts for the gaze target tasks. However, manually designing these prompts may overlook part of the task-relevant factors. As shown in Fig. 4 (right), we choose to prompt the LLM itself to automatically generate these task CoT prompts. Guided by them, the LLM in our model will first analyze the 3D spatial relationship between humans and objects, and then predict the possible gaze targets via mining high-dimensional semantic cues.

Output probability dictionary. In addition to guiding the LLM to perform CoT reasoning step by step, we also require it to output its gaze target predictions in the form of a probability dictionary at the end. This is completed to facilitate batch processing and modal transition of the LLM's predictions. Therefore, in addition to the above task CoT prompts, we also provide the LLM in our model with a format control prompt and a template for the output probability dictionary. The key-value pairs in this dictionary contain the possible gaze targets and their probabilities predicted by the LLM. The output probability dic-

tionary corresponding to each input sample can be defined as D_{out} . Then, we obtain a batch of output dictionaries $D_{\text{out}}^k, k = 1, 2, \dots, K$ corresponding to the batch of input samples $I_k, k = 1, 2, \dots, K$. This approach guarantees that the LLM in our model will consistently provide gaze target predictions in a fixed format.

3.2 Plug-and-play integration mechanism

Modal transition of the LLM's prediction. Although the predictions of the LLM in our zero-shot reasoning module are formatted into a probability dictionary, it is still difficult to directly integrate them with the gaze heatmap generated by existing vision-based gaze target models. Therefore, we generate a multi-hot gaze heatmap for each human $\{o_i \mid i = 1, 2, \dots, M\}$ based on the LLM's multi-hot gaze target predictions. For the human o_i , we set a 2D Gaussian distribution for each possible gaze target $\{o_j \mid j \in \{1, 2, \dots, N\} \cap j \neq i\}$,

$$Gauss_{ij} = p_{ij} \cdot \frac{1}{2\pi\sigma_j^2} \exp\left(-\frac{(x-\mu_j)^T(x-\mu_j)}{2\sigma_j^2}\right), \quad (1)$$

where $\sigma_j^2 = \left(\frac{r_j}{2}\right)^2$. $x = (w, h)$ denotes any point in the image space. r_j is the radius of the possible gaze target o_j , and $\mu = (w_c^j, h_c^j)$ represents the center point of it. The predicted probability p_{ij} of o_j is set as the peak value of the 2D Gaussian distribution. Then, we add up the Gaussian distributions of all possible gaze targets $\{Gauss_{ij} \mid j \in \{1, 2, \dots, N\} \cap j \neq i\}$ in the image space to obtain the multi-hot gaze heatmap M_{llm}^i for the human o_i , and set all heat values greater than 1 to 1,

$$M_{\text{llm}}^i = \min(\sum_{j=1}^N Gauss_{ij}, 1). \quad (2)$$

Gaze heatmap fusion. For the human o_i , we define the single-hot gaze heatmap output by existing vision-based gaze target models as M_{vm}^i . Then, we multiply it directly with the multi-hot gaze heatmap M_{llm}^i output by our LLM reasoning module to obtain the final fusion heatmap M_{fus}^i ,

$$M_{\text{fus}}^i = \text{norm}(M_{\text{llm}}^i + b_{\text{llm}}) \cdot M_{\text{vm}}^i, \quad (3)$$

where b_{llm} denotes the bias added to the heatmap M_{llm}^i to enhance its fault tolerance, and is set to 0.1 through experiments. $\text{norm}(\cdot)$ represents the normalization operation. We use the maximum value point in the fusion heatmap M_{fus}^i as the predicted gaze point of the human o_i , and the object or human corresponding to it as the object-level gaze target. Through this simple cross-modal integration mechanism, our zero-shot LLM reasoning module can be seamlessly integrated with existing vision-based gaze target models in a plug-and-play manner.

4 Experiment

We choose the SOTA LLM ChatGPT o3-mini-high [42] in our LLM reasoning module. Besides, we include multiple open-source gaze target detection models [13, 16, 18, 19, 24], look-at-each-other (LAEO) detection models [15, 71], and shared attention (SA) detection models [13, 51] in our experiments without any additional training or fine-tuning. Among them, Gaze-LLE [24] is the current SOTA gaze target method.

For the gaze target detection task, we evaluate our GazeLLM framework on the most commonly used datasets GazeFollow [10] and VideoAttnTarget [13]. For multi-person social gaze prediction, we evaluate our method in the SA task on the VideoCoAtt dataset [14], and in the LAEO task on the UCO-LAEO and AVA-LAEO datasets [15]. Among them, GazeFollow and VideoAttnTarget only have the gaze point annotations. This is insufficient to evaluate the model performance at the object level. Therefore, we introduce additional object-level ground truths for them. For the test set of GazeFollow dataset, considering that we already detected all LVIS-category objects (see Sect. 3.1), we directly perform position matching between the gaze point labels and these objects. Considering that each test sample contains up to 10 gaze point labels, we treat the objects containing at least 2 gaze points as the object-level gaze target. Finally, we obtain approximately 4200 samples with object-level ground truths out of a total of 4782 test samples.

We follow the standard metrics from existing gaze target benchmarks. L_2 distance measures the distance between the predicted gaze point and the corresponding ground truth. Out-of-frame AP in the VideoAttnTarget benchmark [13] is the average precision for identifying out-of-frame instances. AUC is the area under curve criteria proposed by Judd et al. [72] to assess the confidence of the predicted gaze heatmap.

4.1 Evaluation in gaze target detection

For the proposed GazeLLM framework, we first evaluate the effect of integrating our zero-shot LLM reasoning module with existing vision-based gaze target models in a plug-and-play manner on the standard benchmarks. As shown in Table 1, all existing models achieve substantial performance improvements with the help of our GazeLLM framework. On the GazeFollow benchmark [10], the minimum L_2 distance errors of existing models decrease by between 17% and 25% and the average L_2 distance errors decrease by between 9% and 15%. On the VideoAttnTarget benchmark [13], their L_2 distance errors also decrease by between 11% and 20%. Since the performances of existing vision-based models on AUC and out-of-frame AP metrics already exceed 0.9, the performance improvement brought by our GazeLLM framework is remarkable. As shown in Fig. 5, we also conduct qualitative experiments to demon-

Table 1 Results of evaluating the integration of our zero-shot large language model (LLM) reasoning module with existing vision-based gaze target models in a plug-and-play manner within our GazeLLM framework (“+ GazeLLM”) on the standard benchmarks GazeFollow [10] and VideoAttnTarget [13]. *: the method without temporal module. AP_{oof} : average precision (AP) metric of out-of-frame classification. The best results are highlighted in bold

Methods	GazeFollow			VideoAttnTarget		
	Min. L_2 ↓	Avg. L_2 ↓	AUC ↑	L_2 Dist. ↓	AP_{oof} ↑	AUC ↑
Chong* [13]	0.077	0.137	0.921	0.147	0.848	0.854
Chong* + GazeLLM	0.058	0.116	0.934	0.118	0.883	0.875
Fang [16]	0.067	0.124	0.922	0.108	0.896	0.905
Fang + GazeLLM	0.051	0.108	0.936	0.096	0.911	0.916
Gupta [18]	0.056	0.114	0.943	0.110	0.879	0.913
Gupta + GazeLLM	0.045	0.102	0.952	0.097	0.902	0.922
Miao [19]	0.065	0.123	0.934	0.109	0.908	0.917
Miao + GazeLLM	0.049	0.107	0.945	0.097	0.918	0.925
Gaze-LLE [24]	0.041	0.099	0.958	0.103	0.903	0.937
Gaze-LLE + GazeLLM	0.034	0.091	0.964	0.092	0.915	0.942
Human	0.040	0.096	0.924	0.051	0.925	0.921

Table 2 Evaluation on the extended object-level ground truths in the GazeFollow test set. All data in this table are the average L_2 distance error. “Sports Ball”, “Cell Phone”, “Frisbee”, “Kite”: common activity-related gaze targets. For convenience of presentation, we merge all the ball categories defined in the LVIS category set, e.g., “Basketball”, “Soccer Ball”, and “Tennis Ball”, into a single category called “Sports Ball”. “Long Dist.”: long-distance gaze targets. “Long Tail”: long-tail category gaze targets

Methods	Sports Ball ↓	Cell Phone ↓	Frisbee ↓	Kite ↓	Long Dist. ↓	Long Tail ↓
Chong* [13]	0.114	0.095	0.141	0.178	0.242	0.155
Chong* + GazeLLM	0.084	0.064	0.093	0.126	0.186	0.125
Improvement ratio	26%	33%	34%	29%	23%	19%
Fang [16]	0.097	0.088	0.127	0.163	0.218	0.148
Fang + GazeLLM	0.075	0.063	0.089	0.121	0.174	0.123
Improvement ratio	23%	28%	30%	26%	20%	17%
Gupta [18]	0.090	0.082	0.116	0.151	0.196	0.136
Gupta + GazeLLM	0.071	0.060	0.085	0.116	0.163	0.115
Improvement ratio	21%	27%	27%	23%	17%	15%
Miao [19]	0.096	0.087	0.125	0.160	0.215	0.146
Miao + GazeLLM	0.074	0.062	0.088	0.119	0.171	0.121
Improvement ratio	23%	29%	30%	26%	20%	17%
Gaze-LLE [24]	0.081	0.074	0.101	0.131	0.172	0.119
Gaze-LLE + GazeLLM	0.067	0.057	0.078	0.106	0.148	0.105
Improvement ratio	17%	23%	23%	19%	14%	12%

strate how our zero-shot LLM reasoning module enhances the performance of existing gaze target models.

To thoroughly explore in which cases our GazeLLM framework can bring more substantial improvements to existing gaze target models, we conduct more comprehensive evaluations on the object-level ground truths we extended for the GazeFollow test set. As shown in Table 2, we test on all instances of the following object-level gaze targets: (1) common activity-related gaze targets (“Sports Ball”, “Cell Phone”, “Frisbee”, and “Kite”); (2) long-distance gaze targets (“Long Dist.”) whose L_2 distance from the target center to the human face exceeds 0.5 times of the image’s diagonal length; (3) long-tail category gaze targets

(“Long Tail”) with a frequency of less than 0.2% in the GazeFollow training set.

With the help of our GazeLLM framework, existing vision-based gaze target models achieve a reduction in average L_2 distance error of between 17% and 34% in predicting common activity-related gaze targets. These results demonstrate that our zero-shot LLM reasoning module effectively uncovers the semantic cues between human activities and gaze attention. Besides, existing gaze target models achieve an improvement of between 14% and 23% for long-distance gaze targets. These results demonstrate that our zero-shot LLM reasoning module effectively mitigates the limitations of existing vision-based models in capturing long-range contextual dependencies within im-

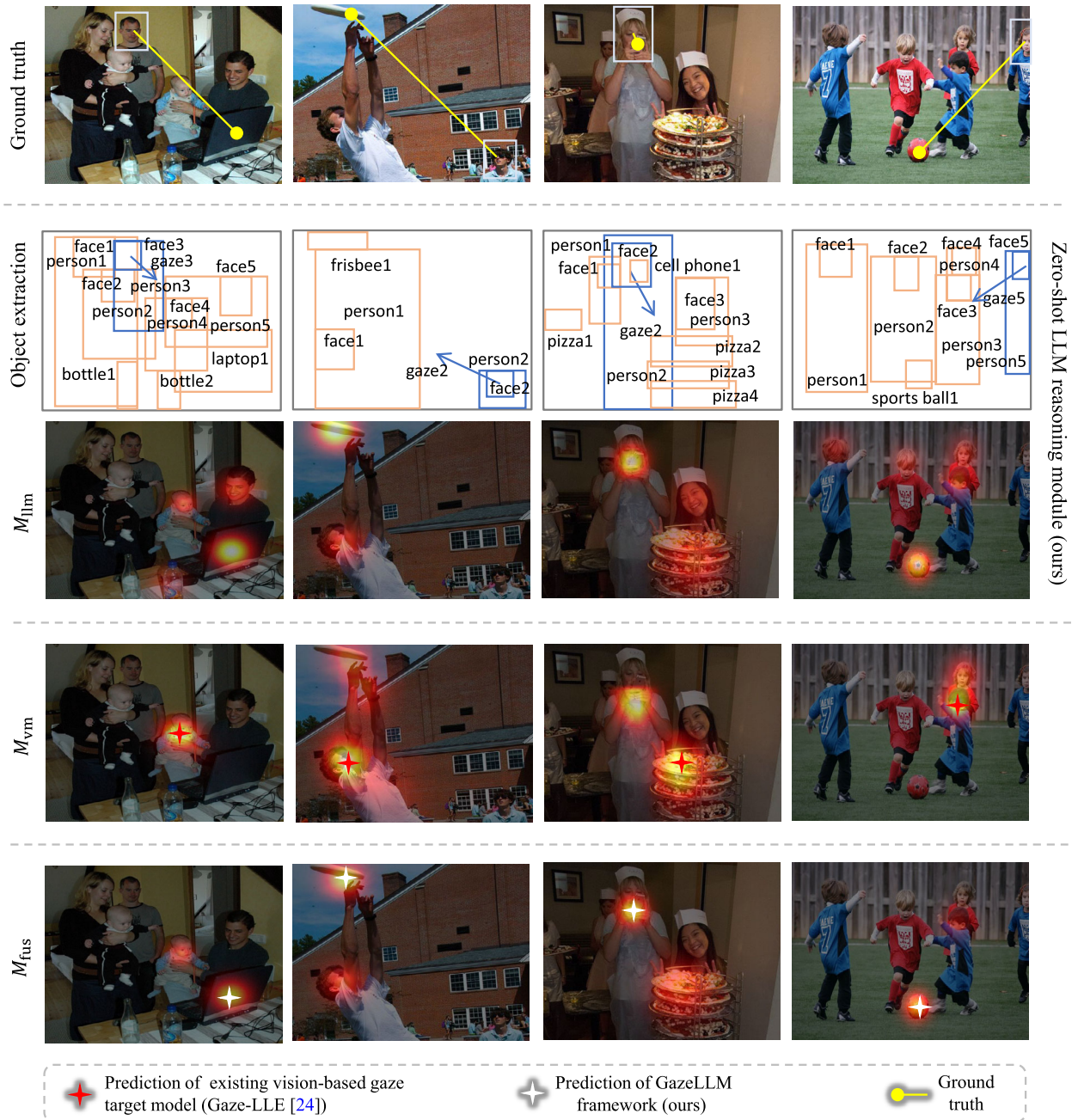


Figure 5 Visualization of how our zero-shot LLM reasoning module enhances the performance of existing vision-based gaze target models. Gaze-LLE [24] is the state-of-the-art vision-based method

ages. Moreover, existing gaze target models also achieve a 12% to 19% improvement in the long-tail category of gaze targets. These results demonstrate that our zero-shot LLM reasoning module effectively mitigates the negative impact of the imbalanced distribution of gaze targets inherent in datasets on existing vision-based models.

4.2 Evaluation in social gaze prediction

We conduct a series of experiments to validate the effectiveness of the proposed GazeLLM framework in multi-person social gaze predictions. As shown in Table 3, we evaluate the performance of integrating our zero-shot LLM reasoning module with existing vision-based shared attention (SA) models in a plug-and-play manner on the

Table 3 Results of evaluating the integration of our zero-shot LLM reasoning module with existing vision-based shared attention (SA) models in a plug-and-play manner on the standard VideoCoAtt benchmark [14]

Methods	VideoCoAtt	
	L_2 Dist. ↓	Accuracy ↑
Chong* [13]	57	83.3
Chong* + GazeLLM	43	92.2
Improvement ratio	25%	11%
HGTTR [51]	46	90.4
HGTTR + GazeLLM	40	94.5
Improvement ratio	13%	5%

Table 4 Results of evaluating the integration of our zero-shot LLM reasoning module with existing vision-based look-at-each-other detection (LAEO) models on the standard AVA-LAEO and UCO-LAEO benchmarks [15]

Methods	AVA-LAEO	UCO-LAEO
	AP ↑	AP ↑
LAEO-Net [15]	50.6	79.5
LAEO-Net + GazeLLM	72.0	91.4
Improvement ratio	42%	15%
LAEO-Net++ [71]	68.7	86.7
LAEO-Net++ + GazeLLM	81.9	95.2
Improvement ratio	19%	10%

standard VideoCoAtt benchmark [14]. The L_2 distance error of the existing models decreases by between 13% and 25%, while the detection accuracy increases by between 5% and 11%. We also evaluate integrating our zero-shot LLM reasoning module with existing vision-based look-at-each-other detection (LAEO) models on the standard AVA-LAEO and UCO-LAEO benchmarks [15]. As shown in Table 4, the LAEO AP metric for existing models increases by between 19% and 42% on the AVA-LAEO benchmark and increases by between 10% and 15% on the UCO-LAEO benchmark. These results demonstrate that our GazeLLM framework exhibits greater flexibility and generalization in understanding various human gaze behaviors compared to traditional vision-based methods.

4.3 Ablation study

To comprehensively evaluate the effectiveness of the proposed GazeLLM framework, we conduct a thorough validation and ablation study of our zero-shot LLM reasoning module on the object-level ground truths that we extended for the GazeFollow test set. Since the LLM in our module performs multi-hot gaze target predictions, we use the common Top-N accuracy metric to evaluate its prediction accuracy and fault tolerance. This metric indicates whether the Top-N most likely predictions contain the ground-truth gaze target. As shown in Table 5, our zero-shot LLM reasoning module achieves a Top-1 accuracy of over 86% and a Top-3 accuracy of over 92% on the

Table 5 Results of evaluating different variants of our zero-shot LLM reasoning module in object-level gaze target prediction on the GazeFollow test set

Methods	GazeFollow	
	Top-1	Top-3
GazeLLM w/o gaze	0.481	0.679
GazeLLM w/o CoT	0.807	0.882
GazeLLM w/o depth	0.824	0.895
GazeLLM	0.863	0.926

Table 6 Results of evaluating the integration of different variants of our zero-shot LLM reasoning module with the state-of-the-art vision-based gaze target model Gaze-LLE [24] on the GazeFollow test set

Methods	GazeFollow		
	Min. L_2 ↓	Avg. L_2 ↓	AUC ↑
Gaze-LLE [24]	0.041	0.099	0.958
Gaze-LLE + Object Heatmap	0.045	0.104	0.951
Gaze-LLE + GazeLLM (w/o gaze)	0.042	0.100	0.956
Gaze-LLE + GazeLLM (w/o CoT)	0.037	0.094	0.962
Gaze-LLE + GazeLLM (w/o depth)	0.036	0.093	0.962
Gaze-LLE + GazeLLM	0.034	0.091	0.964

GazeFollow test set. These results demonstrate that our module has good accuracy in gaze target detection and its multi-hot prediction approach exhibits strong fault tolerance. Besides, we also include experiments of its variants integrated with the SOTA vision-based gaze target model Gaze-LLE [24] in Table 6. Moreover, we visually present the CoT reasoning process of our zero-shot LLM reasoning module in Fig. 6.

We ablate the pretrained LLM in our zero-shot LLM reasoning module and directly generate a multi-hot heatmap for all detected objects. The peak heat value of each object is set to 1. Then, we fuse this object heatmap with the corresponding output of the SOTA vision-based model Gaze-LLE [24], following the same integration mechanism in Sect. 3.2, to obtain the fusion prediction. As shown in Table 6 (“Gaze-LLE + Object Heatmap”), these results demonstrate that simply integrating the extracted object information into existing gaze target models may have a negative impact.

We ablate the gaze direction from the extracted object information in Sect. 3.1, and enable the pretrained LLM to determine possible gaze targets based solely on the remaining object information. As shown in Table 5 (“GazeLLM w/o gaze”) and Table 6 (“Gaze-LLE + GazeLLM (w/o gaze)”), the lack of gaze direction severely limits the accuracy of the LLM in predicting possible gaze targets.

We ablate the task CoT prompts in Sect. 3.1, and use “Which object may the person be looking at?” as the new



Figure 6 Visualization of the CoT reasoning process of our zero-shot LLM reasoning module

prompt to guide the pretrained LLM to directly predict the possible gaze targets in one step. As shown in Table 5 (“GazeLLM w/o CoT”) and Table 6 (“Gaze-LLE + GazeLLM (w/o CoT)”), these results demonstrate that guiding the LLM step by step through CoT prompts for the gaze target tasks can effectively improve its prediction accuracy.

We also ablate the metric depth information in Sect. 3.1, and the pretrained LLM will reason the possible gaze targets only from the extracted 2D object information. As shown in Table 5 (“GazeLLM w/o depth”) and Table 6

(“Gaze-LLE + GazeLLM (w/o depth)”), these results indicate that 3D location information helps the LLM analyze the spatial relationship between humans and objects more accurately, thereby improving its accuracy in gaze target prediction.

We compare the performance of different LLMs in our reasoning module. As shown in Table 7, these results demonstrate that the stronger the reasoning ability of pretrained LLMs, the better the performance of our reasoning module in gaze target prediction.

Table 7 Comparison of different pretrained LLMs in our reasoning module in object-level gaze target prediction on the GazeFollow test set

Methods	Temp.	Seed	GazeFollow	
			Top-1	Top-3
DeepSeek-V3 [40]	0.3	42	0.778	0.864
ChatGPT o3-mini-medium [42]	/	/	0.813	0.889
Qwen3-Max [73]	0.6	42	0.825	0.891
DeepSeek R1 [41]	/	/	0.837	0.907
ChatGPT o3-mini-high [42]	/	/	0.863	0.926

Table 8 Robustness analysis of the impact of object-level input data on our LLM reasoning module. Pert.: perturbation amplitude. None: no perturbation

Methods	Pert.	GazeFollow	
		Top-1	Top-3
Bbox	0 ~ 0.1	0.841	0.905
Gaze	0 ~ 0.1	0.827	0.889
Depth	0 ~ 0.1	0.848	0.912
None	0	0.863	0.926

4.4 Robustness analysis

As described in Sect. 3.1, our GazeLLM module stacks pretrained MM-GroundingDINO [62], GazeOnce [64], and Depth Anything V2 [65]. Errors in object detection, gaze estimation, or metric depth could amplify through the LLM. To evaluate the robustness of our LLM reasoning method, we apply random perturbations to these object-level input data. As shown in Table 8, under random perturbations within the range of 0 to 0.1, the prediction accuracy of the LLM experienced a slight decline. The most significant performance degradation was caused by perturbations to gaze direction. This is probably because the LLM's prediction of potential gaze targets depends on the spatial position relationship between the direction of gaze and the locations of objects.

4.5 Comparison with pretrained VLMs

We compare the zero-shot performance of our LLM reasoning module with that of existing pretrained VLMs in predicting object-level gaze targets. To ensure fairness, we do not provide the involved VLMs with additional training and use the same task CoT prompts as in Sect. 3.1 to guide them in producing the probability in the same way as the proposed method. In this way, the probability can be fused with any existing gaze target models, and we can provide both quantitative and qualitative gains of our LLM reasoning method with VLMs. As shown in Table 9 (quantitative comparison) and Fig. 7 (qualitative comparison), our LLM reasoning module substantially outperforms pretrained VLMs in predicting the possible gaze targets. Existing VLMs are better at descriptive prediction of gaze behavior, but not at accurate detection of gaze object. Thanks

Table 9 Quantitative comparison between our GazeLLM module and pretrained VLMs in object-level gaze target detection

Methods	GazeFollow	
	Top-1	Top-3
InternVL2.5 [74]	0.804	0.882
Qwen2.5-VL-72B [32]	0.816	0.891
GazeLLM	0.863	0.926

to the powerful reasoning ability brought by large-scale reinforcement learning, even without image input, the pretrained LLM in our module can still accurately predict the possible gaze targets from the extracted object information.

4.6 Failure cases

Since our LLM reasoning module relies solely on textual object information, it performs poorly on the cases without clear gaze objects. As shown in Fig. 8, when a person is looking at an occluded object, the ground, wall or sky, the pretrained LLM in our module is unable to provide a correct prediction.

5 Discussion

Although our GazeLLM framework achieves notable improvements in understanding various human gaze behaviors, it still has limitations. The LLM reasoning process introduces additional latency (≈ 300 ms per sample [42]). We mitigate this through batch processing optimizations and note that recent LLM compression techniques could further reduce resource requirements. The energy consumption of LLM inference (≈ 0.02 kWh per 1000 samples [42]) warrants consideration. However, our approach avoids retraining foundation models, presenting a favorable tradeoff compared to fine-tuning VLMs.

By improving gaze target prediction accuracy and interpretability, GazeLLM could benefit assistive technologies (e.g., gaze-controlled interfaces for motor-impaired users), human-robot interaction systems, and driver attention monitoring in autonomous vehicles. The semantic reasoning capability enables more natural explanations of human gaze behaviors. In the future, this technology could help align machine cognition with human cognition, enabling better communication between humans and AI, and contributing to the realization of new human-machine collaboration technology paradigms such as embodied or symbodied AI [75]. While gaze tracking has legitimate applications, malicious use for covert attention monitoring could infringe on personal privacy. We advocate strict adherence to ethical AI guidelines and recommend deployment only in consent-based scenarios.

6 Conclusion

This work presents GazeLLM, a novel framework that leverages pretrained LLMs to overcome the semantic lim-

Ground truth		
Task prompt	Where are the person1 with <code><box>face location</box></code> <code><ref>[0.257, 0.211, 0.404, 0.499]</ref></code> and the person2 with <code><box>face location</box></code> <code><ref>[0.625, 0.159, 0.806, 0.505]</ref></code> looking?	Where are the person1 with <code><box>face location</box></code> <code><ref>[0.070, 0.176, 0.410, 0.412]</ref></code> and the person2 with <code><box>face location</box></code> <code><ref>[0.420, 0.168, 0.787, 0.453]</ref></code> looking?
LLaVA-1.5 [25]	The person on the left is wearing a straw hat and overalls, while the person on the right is wearing a brown hat and a jacket. The person on the left is clapping, while the person on the right is looking down.	One person is using a laptop, and the other person is using a computer. The setting appears to be a conference room or a meeting room.
Qwen2-VL [31]	In the image, the person1 is looking to their right, while the person2 is looking to their left.	The person1 appears to be using a laptop, possibly typing or working on something. The person2 is also using a laptop.
Our LLM reasoning module (zero-shot prediction)	<pre> { Person1's gaze target: {out of frame: 0.9, person2's face: 0.2}, Person2's gaze target: {person1's face: 0.8, person1's body: 0.3}, Look-at-each-other predictions: {[person1, person2]: 0.1}, Shared attention predictions: no instance } </pre>	<pre> { Person1's gaze target: {laptop1: 0.7, laptop2: 0.4, laptop3: 0.2, ...}, Person2's gaze target: {laptop2: 0.8, laptop1: 0.3, person1's face: 0.1, ...}, Look-at-each-other predictions: {[person1, person2]: 0.1}, Shared attention predictions: {[laptop1, person1, person2]: 0.2, [laptop2, person1, person2]: 0.3, [laptop3, person1, person2]: 0.1}, } </pre>

Existing VLMs (zero-shot prediction)

Figure 7 Qualitative comparison of the zero-shot performance between our LLM reasoning module and existing pretrained VLMs in object-level gaze target prediction



Figure 8 Failure cases of our LLM reasoning module. When a person is looking at an occluded object or the ground/wall/sky, the pretrained LLM in our module is unable to provide a correct prediction

itations of vision-based gaze target detection. Our approach is defined by three core contributions. First, we proposed a zero-shot reasoning module. By reformulating gaze detection as a text-based spatial reasoning task, we enable LLMs to predict gaze targets from structured 3D object information, achieving 86.3% Top-1 accuracy on object-level prediction. Second, we introduced a generic integration mechanism. The proposed Gaussian-based heatmap fusion allows plug-and-play enhancement of existing models (e.g., Gaze-LLE), reducing L_2 errors by between 23% and 34% for activity-related targets and setting new SOTAs in social gaze tasks. Third, our approach offers flexibility that is independent of the task. Simply adjusting the CoT prompts enables GazeLLM to be extended to downstream tasks (e.g., shared attention detection), outperforming specialized visual architectures. Despite latency constraints, GazeLLM provides a resource-efficient alternative to VLM fine-tuning. Future work will explore LLM compression for real-time deployment, addressing failure cases, including occluded targets. This framework opens avenues for LLM-powered semantic reasoning in multimodal perception, with potential applications in assistive robotics and human-computer interaction.

Abbreviations

AUC, area under curve; CoT, chain of thought; LAEO, look at each other; LLM, large language model; OOF, out of frame; SA, shared attention; ViT, vision transformer; VLM, vision-language model.

Author contributions

FL performed the study conception. Data collection and method design were performed by YY. FL helped perform the analysis with constructive discussions. The first draft of the manuscript was written by YY. FL revised and approved the final manuscript.

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Data availability

The datasets analyzed in this study, VideoAttnTarget, are available in the repository at <https://github.com/ejcg/attention-target-detection>, and GazeFollow can be accessed at <http://gazefollow.csail.mit.edu/>.

Declarations

Competing interests

The authors have no relevant financial or non-financial interests to disclose.

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